Order-Agnostic Decoding for RNA Pseudoknot Design: Sample-Efficient Generation and Native Motif In-Painting from One Checkpoint

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Abstract

RNA inverse folding, the task of designing a sequence that folds into a target structure, is the computational backbone of synthetic-biology applications from mRNA therapeutics to riboswitch and aptamer engineering. However, state-of-the-art methods rely on best-of-N screening to compensate for low per-sample reliability and left-to-right autoregression, which locks in upstream nucleotides before downstream pairing partners are observed. Here, we introduce an order-agnostic decoder trained under a uniform-permutation distribution over decoding orders, with native motif in-painting from the same checkpoint at inference time. On the OpenKnot Round 7b 240-mer pseudoknot benchmark, our method produces five times as many perfect-structure designs per sample as the strongest published autoregressive baseline, and recovers diversity through order randomization rather than token sampling. To showcase in-painting capability, we perform both motif preservation and motif redesign, the latter strictly impossible under autoregressive decoding. Together, these capabilities enable functional motif redesign for aptamer and riboswitch engineering, and reframe sample-efficient generation as a principled alternative to the dominant best-of-N regime in RNA design.

1. Introduction

RNA secondary structure determines function across ribozymes, viral elements, riboswitches, and aptamers (Damase et al., 2021; Zhang et al., 2023; Wachsmuth et al., 2013; Penchovsky, 2014). Designing a sequence that folds into a target structure—*RNA inverse folding*—is a central computational problem in synthetic biology: redesigning the regulatory scaffold around an aptamer ligand-binding pocket, or engineering a riboswitch around a fixed regulatory motif, both reduce to constrained sequence design under a target topology. Pseudoknotted targets are the hardest sub-class because crossing base pairs couple distant positions.

The dominant evaluation paradigm for data-driven RNA inverse folding is best-of-N screening. State-of-the-art pseudoknot results (He & Sun, 2026) were obtained by drawing $K{=}98\,000$ samples per target on 256 A100 GPUs and post-hoc filtering the top candidates. This protocol is a workaround for a deeper problem: each individual sample from an autoregressive (AR) RNA decoder has low probability of reproducing the target structure, so brute-force sampling is required to find any. The same workaround scales poorly to longer or more constrained design problems, because per-sample success rates drop multiplicatively under additional constraints.

We argue that sample efficiency, not sample volume, is the right axis. Concretely, we replace the L$\rightarrow$R AR decoder of Struct2SeQ (He & Sun, 2026) with an order-agnostic decoder trained under a uniform-permutation distribution over decoding orders, and at inference draw a fresh permutation per sample. The same trained checkpoint then supports three distinct sequence-design tasks without retraining or architectural changes: (i) random-permutation generation of full sequences; (ii) *motif preservation*—fix a structural motif (hairpin, pseudoknot stem) to wild-type and redesign the surrounding scaffold, the RNA analogue of RFdiffusion / ProteinMPNN motif scaffolding (Watson et al., 2023; Dauparas et al., 2022); (iii) *motif redesign*—fix the scaffold and redesign only the motif itself, the natural primitive for aptamer / riboswitch engineering (Wachsmuth et al., 2013; Penchovsky, 2014). Modes (ii) and (iii) are inaccessible under L$\rightarrow$R decoding alone.

Contributions.

- **Order-agnostic RNA decoder.** We replace the LSTM-PE plus causal-conv decoder of Struct2SeQ with a Transformer decoder using a learned relative-position bias keyed on the decoding-order permutation, train under a uniform-permutation distribution over orders,

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Submitted to the 2026 Workshop on Generative and Agentic AI for Biology (ICML 2026). Do not distribute.

and draw a fresh permutation per sample at inference.

- **Native motif in-painting.** A fixed-mask pre-fill of the partner-constraint tensor exposes wild-type nucleotide identities to the structural masks from decode step zero, enabling motif preservation and motif redesign from a single checkpoint without retraining.
- **Controlled architecture-vs-order ablation.** A same-checkpoint comparison on the OpenKnot Round 7b 240-mer pseudoknot benchmark cleanly isolates the per-sample reliability gain from decoding order, independent of architecture and additional training.
- **Mechanistic explanation of rescue asymmetry.** We show that the asymmetric utility of post-hoc local rescue across decoding regimes reduces to failure-mode locality: autoregressive decoders produce over-pair errors that local mutation repairs, while random-permutation decoders produce context-dependent under-pair errors that local repair cannot create.

2. Related work

Modern RNA inverse folding for 2D structure includes ViennaRNA (Lorenz et al., 2011), MCTS-RNA (Yang et al., 2017), RL-based LEARNA (Runge et al., 2019) and RNA-DCGen (Hossain et al., 2024). Struct2SeQ (He & Sun, 2026) provides the strongest published results on OpenKnot Round 7 / 7b pseudoknots via deep Q-learning (Mnih et al., 2015; Sutton & Barto, 2018) on RibonanzaNet-derived rewards (He et al., 2024); we use its DQN training framework, RibonanzaNet-SS reward and Hungarian-Jaccard scoring as shared infrastructure for fair comparison. 3D RNA design (gRNAde (Joshi et al., 2024), transfers of ProteinMPNN / RFdiffusion to RNA backbones) is bottlenecked by 3D-structure data scarcity (Townshend et al., 2021) and underperforms on pseudoknots.

Order-agnostic / any-order autoregressive modelling has been studied in language (Yang et al., 2019; Du et al., 2021; Hoogeboom et al., 2022) and discrete diffusion (Austin et al., 2021; Chang et al., 2022). The biopolymer analogue is ProteinMPNN (Dauparas et al., 2022), whose graph network decodes residues in random order and natively handles partial-sequence constraints. We transport this design lever to RNA secondary-structure design under chemical-mapping-grounded rewards (Lee et al., 2014; He et al., 2024; Wayment-Steele et al., 2022). Best-of-N critiques in the language-model literature (Lightman et al., 2024; Wang et al., 2023) parallel the sample-efficiency framing we adopt.

3. Method

Setup. We adopt the encoder-decoder formulation, RibonanzaNet-SS-derived dense per-position reward, twice-shifted deep-Q-learning training with experience replay, and Hungarian-matched Jaccard scoring of Struct2SeQ (He & Sun, 2026; He et al., 2024; Mnih et al., 2015); we refer the reader there for full details.

Algorithm 1 Random-permutation argmax decoding.

1: **Input:** target structure S, length L, policy π_θ.
2: Sample $\sigma \sim \mathrm{Uniform}(S_L)$.
3: Initialise output $y \leftarrow [\perp]^L$, KV cache $\mathcal{K} \leftarrow \varnothing$.
4: **for** $t = 0, \dots, L-1$ **do**
5: $\quad q \leftarrow \pi_\theta(\sigma_t \mid y, S, \sigma, \mathcal{K})$ {Q-values at position σ_t}
6: $\quad q \leftarrow q + \mathrm{Mask}(\sigma_t, y, S)$ {partner / repeat / A-budget masks}
7: $\quad y_{\sigma_t} \leftarrow \arg\max q$; update $\mathcal{K}$.
8: **end for**
9: **return** y.

Algorithm 2 In-painting via fixed-mask pre-fill.

1: **Input:** target S, length L, fixed mask $f \in (\{\perp\} \cup \{\mathrm{A, U, G, C}\})^L$, policy π_θ.
2: **Pre-fill:** initialise $y \leftarrow f$ at all positions where $f_i \neq \perp$. {Fixed nucleotides visible to partner-mask from step 0}
3: Sample $\sigma \sim \mathrm{Uniform}(S_L)$.
4: **for** $t = 0, \dots, L-1$ **do**
5: $\quad$ Decode σ_t as in Algorithm 1, line 5–6.
6: $\quad$ **if** $f_{\sigma_t} \neq \perp$ **then**
7: $\quad\quad y_{\sigma_t} \leftarrow f_{\sigma_t}$ {teacher-force; KV cache stays consistent}
8: $\quad$ **else**
9: $\quad\quad y_{\sigma_t} \leftarrow \arg\max q$.
10: $\quad$ **end if**
11: $\quad$ Update $\mathcal{K}$.
12: **end for**
13: **return** y.

Architecture and training. The encoder follows He & Sun (2026) unchanged. We replace the LSTM-PE plus causal-conv decoder with a pure Transformer decoder using a learned relative-position bias on the self- and cross-attention scores, keyed on the decoding-order permutation σ. Position information is no longer baked into a recurrent state but supplied to attention scores on the fly, which is what lets arbitrary σ share the same parameters. For each training example of length L we draw $\sigma \sim \mathrm{Uniform}(S_L)$; the causal mask is in step order, not RNA-position order. This realises the order-agnostic training distribution suggested in He & Sun (2026, §4) and is related to permutation-language modelling (Yang et al., 2019) and absorbing-state discrete diffusion (Austin et al., 2021; Hoogeboom et al., 2022). We train for 5 episodes beyond the released Struct2SeQ checkpoint.

Native in-painting vs. teacher-forced AR. The pre-fill in Algorithm 2 (line 2) is the load-bearing construction. Because constrained nucleotides are written into the output buffer *before* the decode loop starts, the partner-mask at every subsequent step reads the constrained identities regardless of where they fall in σ: the model is structurally aware of the motif from step 0. Setting f to wild-type at motif positions yields *motif preservation*; setting f to wild-type at scaffold positions yields *motif redesign*.

By contrast, an L→R AR decoder with teacher-forced motif positions exhibits a position-dependent asymmetry. For a motif at indices $[a, b]$ in a length-L target: at decode steps $t < a$, the model emits scaffold tokens with no information about the motif identity (it has not yet been visited); at $t \in [a, b]$, the emission is overridden to wild-type and enters the KV cache; for $t > b$, downstream scaffold positions are motif-aware. The upstream scaffold is generated *blind* to the motif identity, while the downstream scaffold is *informed*. For pseudoknotted targets where long-range crossing pairs span the motif boundary, upstream commits before its eventual partner's context exists. This is a degraded hack, not a native capability; Table 1 quantifies the empirical 2.3× gap.

4. Experiments

Benchmark. We evaluate on the 20 OpenKnot Round 7b 240-mer pseudoknot puzzles (Eterna Game, 2024; Lee et al., 2014), the longer-and-harder counterpart to OK7 used by He & Sun (2026). Each method generates $K{\sim}1000$ samples per target. Sequences are scored identically across all rows: predicted secondary structure from RibonanzaNet-SS (He et al., 2024) → Hungarian-matched base-pair set → Jaccard against the target. We report perfect-Jaccard rate (fraction of samples with Jaccard $= 1.0$) and puzzles solved (≥ 1 perfect at $K{=}1000$) on every row.

Headline results. Table 1 reports the main comparison. At matched $K{\sim}1000$, our random-permutation argmax decoder produces 33.24% perfect designs and solves 13/20 puzzles. The strongest published AR protocol (3-strategy + rescue, faithful to `test_240.py`) reaches 6.64% / 17/20: our per-sample reliability is 5.0× higher; AR's coverage lead of 4 puzzles comes entirely from rescue (§5). Under matched protocol (both methods with 3-strategy + rescue), the per-sample ratio is 2.0× in our favour and AR leads coverage by 2 puzzles.

Architecture vs. decoding order. On the same checkpoint, L→R + argmax reaches 23.72% / 10 and random-perm + argmax reaches 33.24% / 13 (+9.52 pp from order alone, weights and architecture identical). Holding decoding order fixed at L→R + argmax, the original 10-episode checkpoint reaches 14.35% / 5 versus our 23.72% / 10 (+9.37 pp from architecture plus 5 extra training episodes; confounded). Order and architecture+training each contribute roughly half of the total perfect-rate gain.

Per-strategy AR ablation. The 3-strategy protocol at $K{=}333$ each: for original Struct2SeQ AR, $\epsilon{=}0.05$ achieves 10.77% / 12, $\epsilon{=}0.10$ achieves 8.79% / 11, Q-softmax achieves 0.07% / 2. Our random-perm checkpoint mirrors the pattern: 24.03% / 13, 16.89% / 14, 0.07% / 3. Q-softmax is uncorrelated with correctness because Q-values are not calibrated probabilities (He & Sun, 2026, §2.7); softmax over them drifts off-peak. At $K{=}98\,000$, 0.07% still gives ∼70 perfects per target; at $K{=}333$ it gives ∼0.2.

Diversity engine. L→R + argmax is effectively deterministic at $K{=}1000$: original Struct2SeQ produces 340 unique sequences across 20,032 samples (∼17 per target), and our checkpoint forced into L→R + argmax produces 346 — the remaining ∼98% are exact duplicates from CUDA float non-determinism. Random-permutation + argmax produces 20,032 unique sequences from the same budget, one per sample. Order randomisation, not token sampling, is the diversity engine, and composes with argmax instead of substituting for it.

In-painting. For each target we extract a structural-element inventory—hairpins, internal loops, multi-loops, and pseudoknot stems—via a stem-grouping algorithm built on OpenKnotScorePipeline's pair-crossing classification (**?**), and per sample draw one motif uniformly at random from the inventory (sizes capped at 50% of L, mean selected size ∼15%). Native motif preservation under this protocol reaches 25.69% / 13 at $K{=}1000$; motif redesign—fixing the ∼85% scaffold to wild-type and regenerating only the motif—reaches 19.82% / 13 on the same inventory. Constraint satisfaction is verified 100% on a 500-sequence sample. Motif redesign is inaccessible under L→R AR even with teacher-forcing, because the decoder commits scaffold positions before reaching the motif. Against an AR teacher-forced motif baseline, ours has a 2.3× perfect-rate ratio on motif preservation; the L→R + teacher-forced ablation rows in Table 1 are in flight.

5. Discussion

Per-sample reliability vs. coverage Pareto. The two metrics measure different things and the field has historically conflated them. Best-of-N at $K{=}98\,000$ optimises *coverage*: with enough samples, even a low-reliability decoder will eventually produce one perfect design per target. Our random-perm checkpoint optimises *per-sample reliability*: 33.24% of single samples are perfect at $K{=}1000$, with no rescue and no sampling temperature. These are distinct operating points on a Pareto frontier; our method offers both extremes from one checkpoint (reliability-optimised: 33.24% / 13; coverage-optimised: 13.44% / 15). AR has only one regime under matched compute, because argmax-only is essentially deterministic so coverage requires the full sampling protocol. Per-sample reliability is the axis that

Table 1. OpenKnot Round 7b 240-mer pseudoknot design at matched $K\sim1000$ samples per target. Perfect-Jaccard % is the fraction of designs reproducing the target exactly (Hungarian on RibonanzaNet-SS predicted base-pair probabilities, $\theta=0.5$). Puzzles solved is the count of targets with ≥ 1 perfect-Jaccard design at $K=1000$. The original Struct2SeQ row uses the released checkpoint trained for 10 episodes; "Ours" uses our order-agnostic Transformer-decoder variant trained for 5 additional episodes under the random-permutation distribution. The 3-strategy protocol is ϵ-greedy with $p=0.05$, ϵ-greedy with $p=0.10$, and Q-softmax sampling at $p=1.0$, each with $K\approx333$, matching the `test_240.py` protocol of He & Sun (2026). Rescue is the post-hoc 4^k local repair on the per-puzzle best non-perfect seed when $|\text{diff\_pos}| \leq 4$.

Method	Perfect-Jaccard (%)	Puzzles solved
Plain generation. Original Struct2SeQ AR (L→R, 10 episodes)		
argmax-only	14.35	5/20
3-strategy	6.55	13/20
3-strategy + rescue	6.64	17/20
Plain generation. Ours (bidirectional, 10 + 5 episodes)		
L→R argmax-only	23.72	10/20
L→R + 3-strategy	9.38	12/20
L→R + 3-strategy + rescue	9.13	13/20
Random-perm argmax-only	**33.24**	**13/20**
Random-perm + 3-strategy	13.65	14/20
Random-perm + 3-strategy + rescue	13.44	15/20
In-painting. Ours (random-perm) and AR teacher-forced reference		
Motif preservation	**25.69**	**13/20**
Motif preservation + rescue	25.28	13/20
Motif redesign (AR-impossible natively)	19.82	13/20
In-painting. Ours L→R + teacher-forced motif (architecture vs. order ablation)		
L→R + teacher-forced motif (argmax)	—	—
L→R + teacher-forced motif + 3-strategy	—	—
L→R + teacher-forced motif + 3-strategy + rescue	—	—
L→R + teacher-forced motif (argmax) + rescue	—	—

determines wet-lab synthesis cost and constrained-design feasibility.

Order randomisation as a substitute diversity engine. The published 3-strategy AR protocol (ϵ-greedy at two values plus Q-softmax) is in retrospect a workaround for L→R + argmax determinism. Order randomisation supplies the same diversity through a different mechanism: distinct σ's induce distinct conditioning paths through the same trained policy, producing 1000 distinct argmax outputs from $K=1000$ samples. This is methodologically attractive because it is orthogonal to per-sample reliability (argmax preserves point-estimate quality) rather than substituting for it (ϵ-greedy and Q-softmax inject noise that lowers per-sample reliability while raising diversity). Order is a free parameter the L→R AR decoder cannot exploit.

Why rescue is one-sided: over-pair vs. under-pair failures. Symmetric rescue (4^k local enumeration on the per-puzzle best non-perfect seed) reaches Jaccard $= 1.0$ on $4/5$ AR seeds but $0/6$ random-perm seeds. This is failure-mode asymmetry, not algorithmic asymmetry. AR's narrow exploration produces *over-pair* errors (an extra predicted pair the target lacks), which rescue removes by breaking Watson-Crick complementarity. Random-perm produces *under-pair* errors: the target wants a pair the network refuses to predict from the surrounding $\sim$200 bases of context, and no local mutation creates one. Manual verification on puzzle 12 confirms all 16 mutations of the relevant C-G pair fail (max Jaccard 0.984); multi-seed model-guided in-paint rescue (top-5 seeds, $K=100$) hits the same ceiling. Rescue is a real tool—it adds $+1$ puzzle to our 3-strategy run—but its utility tracks failure-mode locality, which is set by decoding regime.

6. Limitations and conclusion

Limitations. All numbers come from a single evaluation seed; between-config statistical significance is not quantified. Evaluation is computational throughout (RibonanzaNet-SS predicted structure for Jaccard, RibonanzaNet predicted SHAPE for OK score); wet-lab validation is future work. The architecture-vs-order attribution is partially confounded by 5 extra training episodes on our checkpoint, though the same-checkpoint L→R-vs-random-perm gap ($+9.52$ pp) is clean. Our random-perm decoder's failure modes on unsolved puzzles are under-pair errors that cannot be fixed by local repair; closing this seed bottleneck (multi-seed or near-miss-targeted RL fine-tuning) is the most natural follow-up.

Conclusion and future work. Order-agnostic decoding delivers higher per-sample reliability and a strictly larger capability set than L→R AR for RNA pseudoknot design, from a single checkpoint. The natural domain follow-up is aptamer and riboswitch redesign (Wachsmuth et al., 2013; Penchovsky, 2014; Wayment-Steele et al., 2022): fix the ligand-binding aptamer pocket as the constrained motif and redesign the surrounding regulatory scaffold to vary ligand response threshold or kinetics, then validate experimentally. Other high-leverage directions are in-painting-aware training (random K-of-WT mask during training to close the motif-redesign vs. preservation gap), multi-seed and iterated rescue for under-pair failures, and SHAPE-supervised retraining as a Struct2SeQ-SHAPE analogue.

Impact Statement

This work targets RNA inverse folding for pseudoknot-bearing targets, with downstream relevance to mRNA therapeutic design (Damase et al., 2021; Zhang et al., 2023), riboswitch and aptamer engineering (Wachsmuth et al., 2013; Penchovsky, 2014; Wayment-Steele et al., 2022), and synthetic-biology circuit construction. We choose 2D secondary-structure design because chemical-mapping experiments make experimentally-grounded structural feedback abundant in a way 3D crystallographic data is not (Lee et al., 2014; He et al., 2024), allowing the resulting neural oracles to be incorporated directly into the design loop. By raising per-sample reliability and lowering the compute requirement for high-quality candidates ($\sim$100$\times$ less per target than the published Struct2SeQ baseline at matched coverage), the method reduces both compute and wet-lab cost, lowering the barrier for academic labs to participate in RNA design challenges. We see no immediate dual-use concern beyond those generic to nucleic-acid design tools; downstream synthesis is gated by physical reagent access. Interactive deployment on platforms such as Eterna (Lee et al., 2014; Wayment-Steele et al., 2022) is a promising longer-term direction: order-agnostic decoding is a natural fit for a setting in which a human fixes an arbitrary subset of nucleotides and the model fills in the rest in real time, placing the design tool in a transparent collaborative role rather than as an opaque generative oracle.

References

Austin, J., Johnson, D. D., Ho, J., Tarlow, D., and van den Berg, R. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

Chang, H., Zhang, H., Jiang, L., Liu, C., and Freeman, W. T. MaskGIT: Masked generative image transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

Damase, T. R., Sukhovershin, R., Boada, C., Taraballi, F., Pettigrew, R. I., and Cooke, J. P. The limitless future of RNA therapeutics. *Frontiers in Bioengineering and Biotechnology*, 9:628137, 2021. doi: 10.3389/fbioe.2021.628137.

Dauparas, J., Anishchenko, I., Bennett, N., others, and Baker, D. Robust deep learning–based protein sequence design using ProteinMPNN. *Science*, 378(6615):49–56, 2022. doi: 10.1126/science.add2187.

Du, C., Tu, Z., and Jiang, J. Order-agnostic cross entropy for non-autoregressive machine translation. In *International Conference on Machine Learning (ICML)*, 2021. URL https://proceedings.mlr.press/v139/du21c.html.

Eterna Game. OpenKnot AI Design Data. https://github.com/eternagame/OpenKnotAIDesignData, 2024. GitHub repository.

He, S. and Sun, Q. Struct2SeQ: RNA inverse folding with deep Q-learning. *bioRxiv*, 2026. doi: 10.64898/2026.01.16.700031.

He, S., Huang, R., Townley, J., Quarmby, C., others, Wayment-Steele, H. K., and Das, R. Ribonanza: deep learning of RNA structure through dual crowdsourcing. *bioRxiv*, 2024. doi: 10.1101/2024.02.24.581671.

Hoogeboom, E., Gritsenko, A. A., Bastings, J., Poole, B., van den Berg, R., and Salimans, T. Autoregressive diffusion models. In *International Conference on Learning Representations (ICLR)*, 2022. URL https://openreview.net/forum?id=Lm8T39vLDTE.

Hossain, H. S. et al. RNA-DCGen: Dual constrained RNA sequence generation with LLM-attack. *bioRxiv*, 2024. doi: 10.1101/2024.09.23.614570.

Joshi, C. K., Jamasb, A. R., Viñas, R., Harris, C., Mathis, S. V., Morehead, A., Anand, R., and Liò, P. grnade: Geometric deep learning for 3D RNA inverse design. In *ICLR 2024 Generative and Experimental Perspectives for Biomolecular Design Workshop*, 2024.

Lee, J., Kladwang, W., Lee, M., Cantu, D., others, Treuille, A., and Das, R. RNA design rules from a massive open laboratory. *Proceedings of the National Academy of Sciences*, 111(6):2122–2127, 2014. doi: 10.1073/pnas.1313039111.

Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., Leike, J., Schulman, J., Sutskever, I., and Cobbe, K. Let's verify step by step. In *International Conference on Learning Representations (ICLR)*, 2024.

Lorenz, R., Bernhart, S. H., Höner zu Siederdissen, C., Tafer, H., Flamm, C., Stadler, P. F., and Hofacker, I. L. ViennaRNA package 2.0. *Algorithms for Molecular Biology*, 6:26, 2011. doi: 10.1186/1748-7188-6-26.

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., others, and Hassabis, D. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015. doi: 10.1038/nature14236.

Penchovsky, R. Computational design of allosteric ribozymes as molecular biosensors. *Biotechnology Advances*, 32(5):1015–1027, 2014. doi: 10.1016/j.biotechadv.2014.05.005.

Runge, F., Stoll, D., Falkner, S., and Hutter, F. Learning to design RNA. In *International Conference on Learning Representations (ICLR)*, 2019. URL https://openreview.net/forum?id=ByfyHh05tQ.

Sutton, R. S. and Barto, A. G. *Reinforcement Learning: An Introduction*. MIT Press, Cambridge, MA, 2nd edition, 2018.

Townshend, R. J. L., Eismann, S., Watkins, A. M., Rangan, R., Karelina, M., Das, R., and Dror, R. O. Geometric deep learning of RNA structure. *Science*, 373(6558): 1047–1051, 2021. doi: 10.1126/science.abe5650.

Wachsmuth, M., Findeiß, S., Weissheimer, N., Stadler, P. F., and Mörl, M. De novo design of a synthetic riboswitch that regulates transcription termination. *Nucleic Acids Research*, 41(4):2541–2551, 2013. doi: 10.1093/nar/gks1330.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., and Zhou, D. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

Watson, J. L., Juergens, D., Bennett, N. R., others, and Baker, D. De novo design of protein structure and function with RFdiffusion. *Nature*, 620(7976):1089–1100, 2023. doi: 10.1038/s41586-023-06415-8.

Wayment-Steele, H. K., Wu, M., Gotrik, M., and Das, R. Crowdsourced RNA design discovers diverse, reversible, efficient, self-contained molecular switches. *Proceedings of the National Academy of Sciences*, 119(18): e2112979119, 2022. doi: 10.1073/pnas.2112979119.

Yang, X., Yoshizoe, K., Taneda, A., and Tsuda, K. RNA inverse folding using Monte Carlo tree search. *BMC Bioinformatics*, 18(1):468, 2017. doi: 10.1186/s12859-017-1882-7.

Yang, Z., Dai, Z., Yang, Y., Carbonell, J., Salakhutdinov, R., and Le, Q. V. XLNet: Generalized autoregressive pretraining for language understanding. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Zhang, H., Zhang, L., Lin, A., Xu, C., Li, Z., Liu, K., others, Mathews, D. H., and Huang, L. Algorithm for optimized mRNA design improves stability and immunogenicity. *Nature*, 621(7978):396–403, 2023. doi: 10.1038/s41586-023-06127-z.